



POURIA SAEDI

EMBEDDED ENGINEER

CONTACT

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-  LinkedIn
-  Portfolio
-  Milan, Italy

PROFILE

Master's degree in Electrical Engineering specialized in hardware, firmware, and embedded systems. 2+ years professional experience in embedded systems design, PCB development, and IoT solutions.



WORK EXPERIENCE

Embedded Systems Engineer

Self-Employed, Milan | Apr 2025 - Present
Designing network of smart devices with mesh networking and IoT. Implementing AI features and voice interfaces. Engineering modular add-ons ([Smart Home](#)).

Embedded Systems Engineer

OTTI s.r.l.s, Milan | May 2024 - Feb 2025
Developed modular system for boat panel modernization ([Smart Panel](#)). Integrated ESP32, GSM, RTC, and NMEA. Designed compact PCBs.

Embedded Systems Engineer

Toos Bio Research Co., Mashhad | Jan 2022 - Sep 2022
Upgraded stereotaxic system and designed lab device controls. Developed motorized system for eye tracker ([Lab Equipments](#)).

Electronics Engineer Intern

Bartar Andishan Co., Mashhad | Jun 2019 - Aug 2019
Gained hands-on experience in electronics and embedded systems development.

EDUCATION

MS in Electrical Engineering

Politecnico di Milano | Sep 2022 - Apr 2025
Specialized in hardware and embedded systems. Developed expertise in firmware design, PCB development, and system integration.

BS in Electrical Engineering

Ferdowsi University | Sep 2016 - Sep 2021
Foundation in electrical engineering principles. Completed coursework in electronics, embedded systems, and control systems.

TOOLS & FRAMEWORKS

Programming Languages: C, C++, Python
Hardware Description Languages: VHDL, Verilog
PCB design software: EasyEDA, Altium Designer
PLC: Siemens SIMATIC STEP 7, Siemens LOGO, Delta PLC
Frameworks: RTOS, MATLAB, SolidWorks, LabView, PSpice, Proteus
Tools: Git, Excel, $\LaTeX$, Miro, VS Code, ModelSim, VIVADO, Quartus, Arduino IDE, Keil, STM32Cube

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SKILLS

Firmware & Embedded

ESP-IDF, FreeRTOS, RTOS
Keil uVision, STM32CubeIDE
Arduino IDE, BLE Mesh
LVGL, IEEE 1588 PTP

Hardware & PCB

EasyEDA, Altium Designer
PSpice, Proteus
Analog & Digital Design
Sensor & Actuator Integration

FPGA & HDL

Verilog, VHDL
ModelSim, Quartus II
Xilinx ISE, VIVADO

Software & Frameworks

C, C++, Python, Flutter
Django, SQLite, Redis
MATLAB, Simulink
SolidWorks Electrical
Git, L^AT_EX, VS Code

Industrial Automation

Siemens STEP 7, LOGO!Soft
Delta PLC

LANGUAGES

English	Professional
Persian	Native
Italian	Elementary



PROJECTS

Smart Alarm System STM32 · Flutter · EasyEDA · PSpice
Custom alarm PCB with STM32 state-machine firmware (zone management, tamper detection) and Flutter mobile app for re-mote arm/disarm and push notifications.

IEEE 1588 PTP on STM32 STM32 · CAN Bus · Keil uVision
Implemented PTP master/slave state machines over CAN bus for sub-microsecond clock synchronisation; hardware-timer times-tamping to minimise software jitter.

Pipelined MIPS Processor Verilog · Quartus II · ModelSim
Synthesisable 5-stage MIPS pipeline in Verilog with full hazard detection and forwarding; verified in ModelSim and deployed on FPGA via Quartus II.

Neural Network on STM32 Python · Keras · TensorFlow · STM32
Trained, int8-quantised, and deployed a Keras/TensorFlow classifier onto STM32 flash; validated on-device inference accuracy and latency against the Python baseline.

HDL Digital Modules Library VHDL · Verilog · ModelSim · Xilinx ISE
Reusable FIFO, UART, SPI/I2C, PWM, and FSM modules in VHDL and Verilog; verified with ModelSim testbenches and synthesised for Xilinx FPGA.

P10 LED Display Reverse Engineering EasyEDA · SIM800 · Proteus
Reverse-engineered P10 matrix scan-line protocol via logic analyser; built custom controller PCB with SIM800 GSM module for SMS-driven dynamic content display.

AlienFighter Game on STM32 STM32 · Embedded C · Keil uVision
Bare-metal arcade game on STM32 with SPI display, interrupt-driven GPIO/ADC input handling, and a rendering pipeline optimised within MCU RAM/flash constraints.

RF Transmission System Embedded C · EasyEDA · Proteus
Designed transmitter/receiver PCBs with RF front-end matching and power filtering; firmware implements packet framing and error detection, pre-validated in Proteus.

PLC Automation Siemens STEP 7 · LOGO!Soft · Delta PLC
Programmed ladder-logic and FBD sequences for sequential process control including motor interlocks, timer/counter logic, and alarm management across Siemens S7 and Delta platforms.

Power System Load Flow MATLAB · Simulink
Implemented Newton-Raphson and Gauss-Seidel load flow algorithms in MATLAB; modelled multi-bus networks in Simulink for fault analysis and voltage profile studies.

Audio Amplifier Design PSpice · EasyEDA · Proteus
Multi-stage amplifier (pre-amp, tone control, power stage) with bias network and thermal compensation; AC, transient, and THD simulations run before PCB fabrication.